CSE 430

Compiler Design Lab

Lab Work 4

Submission Date: **25/04/2026**

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Session: **Fall 2025**	**University of Asia Pacific**

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Code Link

[Remove Left Recursion]

Removing Direct Left Recursion

Input

E → E+T | T

Output

```
PS D:\Study\(4 - 2) four_Two\CSE 430 Compiler Design Lab\Lab Work - 4>
Original Grammar:
E -> E+T | T

Grammar after removing Left Recursion:
E -> TE'
E' -> +TE' | ε
```

Input

T → T*F | F

Output

```
PS D:\Study\(4 - 2) four_Two\CSE 430 Compiler Design Lab\Lab Work - 4>
Original Grammar:
T -> T*F | F

Grammar after removing Left Recursion:
T -> FT'
T' -> *FT' | ε
```

Input

A → Ac | Aad | bd | c

Output

```
PS D:\Study\(4 - 2) four_Two\CSE 430 Compiler Design Lab\Lab Work - 4>
Original Grammar:
A -> Ac | Aad | bd | c

Grammar after removing Left Recursion:
A -> bdA' | cA'
A' -> cA' | adA' | ε
```

Removing Indirect Left Recursion

Input

S → x | y
A → Sa | Bf | m
B → Ab | Sc | n

Output

```
PS D:\Study\(4 - 2) four_Two\CSE 430 Compiler Design Lab\Lab Work - 4>
Original Grammar:
S -> x | y
A -> Sa | Bf | m
B -> Ab | Sc | n

Grammar after removing Left Recursion:
S -> x | y
A -> xa | ya | Bf | m
B -> xabB' | yabB' | mbB' | xcB' | ycB' | nB'
B' -> fbB' | ε
```

Input

A → Ba | d
B → Cb | g
C → Ac | f

Output

```
PS D:\Study\(4 - 2) four_Two\CSE 430 Compiler Design Lab\Lab Work - 4>
Original Grammar:
A -> Ba | d
B -> Cb | g
C -> Ac | f

Grammar after removing Left Recursion:
A -> Ba | d
B -> Cb | g
C -> gacC' | dcC' | fC'
C' -> bacC' | ε
```